

ECE 564/565, Spring 2026, Fundamentals of Autonomous Robots Lab.

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ECE 564/565 Final Robotic Contest: Robotic Transporters in Warehouses

Mock Contest: April 10th, Friday, 12:30-3:00pm, W.S. Robotics Lab.
Actual Contest: TBA

Equipment Return: Setup a time with the TA.
Report due: TBA, the pdf file & IC-codes on BB.

You must return the equipment completely, in working condition and as organized in the boxes as you received. Otherwise, you will receive no credit from the course.

Overview:

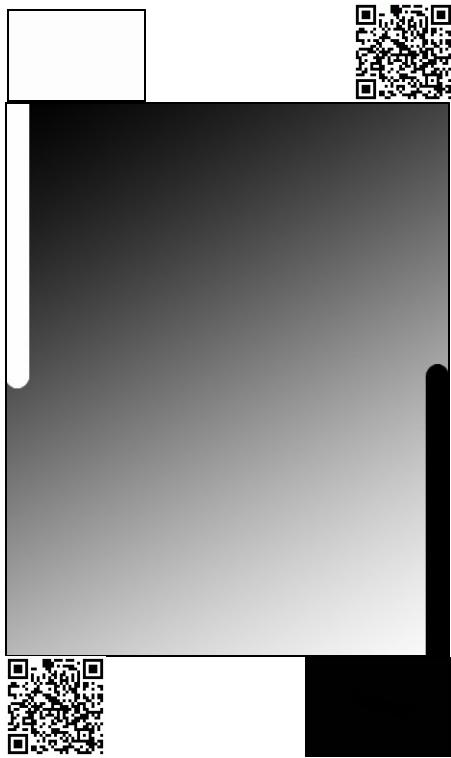
The robotic carriers in warehouses contest mimics the concept of autonomous mobile robots transporting payloads throughout a warehouse without any supervision. For example, Amazon acquired Kiva Systems. This goods-to-person (or location) system uses low-profile mobile robotic vehicles (guided by QR codes on the floor) to drive underneath mobile racks and deliver them to workstations.

The final robotic contest involves two robots and 9 colored cubes of size about **2-inch**. There will be **3 green, 3 purple and 3 yellow**-colored boxes. The goal of the contest is to collect the assigned colored boxes and delivering to the home location in a 2-minute game (special rules apply). The final robotic contest will count for 20% of your grade. Each team can receive extra credits depending on the number of the points they collect.

Rules:

1) Boxes: 9 boxes, about **2-inch cubes**, colored as **green, purple and yellow** (3 boxes from each color) will be randomly distributed on the play arena. No boxes will be placed in the “home locations” at the beginning of each game.

2) Play Arena: The Play Arena is approximately *4x6 feet* rectangular arena with “home locations” at both ends of the arena. The arena will be placed on a table during the competition. The walls of the arena are transparent and approximately 6 inches high. The floor of the arena is a gray color gradient, which is pure white at one home location and going to pure black on the other home location. Each home location is about 1.5 foot wide by 1 foot deep. The home location on the black corner of the arena is white and the other home location on the white corner of the arena is black. Therefore, each team can recognize the “home locations” as shown in the figure below (without the black and white strips in front of the home locations). There will be also two different QR code images on the corners of the table, one for each team. The one along the white home location is belong to the team located on the white home location area and vice versa. These QR images (unknown before each game) will represent green, purple and yellow colors which will be placed on their locations right before each game. The teams will not know in advance which color of boxes are assigned to them to pick and carry.



3) Robots:

- Teams can design any type of robot they like using **only** the provided LEGO Mindstorm Robotic Invention System 2.0 kit.
- Robot must measure at most 1 ft length x 1 ft width.
- There is no limit to the vertical dimension, but robots must fit within a 1 ft square column.
- Each group must design, build, and debug its own unique robot.
- Robot must be a single, physically rigid entity. “Multiple robot” designs are *not* allowed. Intentional “falling parts” from the robots are *not* allowed.
- Teams will be provided with one Wallaby controller, several sensors (including one sonar sensor and one camera for each team) and 2 geared motors which they have used during regular labs. Upon request one additional motor can be provided.
- Teams are **not** allowed to buy and use any other sensors, motors, LEGO pieces than the ones provided. If any team needs any additional sensors, motors, metal brackets and/or lego pieces they must ask the instructor (or TA) for these additional parts. If there are enough additional equipment for all groups, then additional required equipment will be provided to the teams.
- Any types of glues, noxious liquids, tapes, etc. are **not** allowed.
- **TESTING IS VERY IMPORTANT!** After you complete your design, **test your robot against any possible scenarios** that you might think of, getting stuck in the corner, tangling with the opponent robot, blocked with the opponent robot, failing sensors, ambient light conditions (will be different in the room where the actual competition is going to held), selections of tires, speed of the robot, etc.

4) Contest Play:

a) Tournament:

- Tournament number of games and/or elimination rules will be decided later on depending on the result of the Mock competition.
- In the first round of the tournament, robots will be matched to pair more-successful robots from the mock contest against less-successful ones.
- Each game will last 2 minutes.
- Two robots will compete against one another at each game.
- Robots will be manually started on the judge's cue.
 - You must start the robot in this way in the competition:
 - Put your robot on the table in front of its own home location, turn the Wallaby on (robot does not move), and when start signal is given, push the start or stop button **ONCE** to start your robot.
 - Robots may **autonomously** run some calibration code while waiting to start. However, **you** should **NOT** adjust/sent any parameters with pressing start/stop button once or multiple times! **This is NOT allowed.**
- Robots **MUST** automatically stop moving at the end of the 2-minute game or they will be disqualified from that game.
- When the teams are called, they will have 30 seconds to put their robots on the field. Otherwise, they will be disqualified from that game.

b) Mock Contest:

- Robots will play individually for the mock contest. Teams will have two chances (two different colors will be assigned for each run).
- During the first run there will be only 3 boxes (each with different colors) and during the second run there will be 6 boxes (2 from each color).
- Robots must play a valid game and should achieve the following tasks for the mock contest:
 1. Starting the game after the start button is pressed (10 Points)
 2. Traveling to QR code area and learning the assigned color (20 Points)
 3. Distinguishing assigned color boxes from the others (10 Points)
 4. Collecting at least one assigned color box (30 points)
 5. Delivering it to the correct home location (20 Points)
 6. Stopping automatically at the end of 2-mins. (10 Points)
- Related actual contest rules apply to the mock contest, as well.
- **You need to receive at least 75 points to qualify for the final competition.**

c) Actual Contest:

- Actual contest place will be announced.
- Robots may **not** intentionally damage or destroy their opponent.
- Bumping and blocking is allowed.
- **Uncharged batteries are not an excuse!**
- At the beginning of each game, robots will be given a random home location. Robots will start at the orientation of 12 o'clock right in front of their assigned home locations. 12 o'clock is facing upward toward the opponent side.
- Once the actual tournament begins, modifications to a robot's design (including reprogramming) are allowed. However, once a team is called, the team's robot must be in the play arena in 30 seconds. Otherwise, the team will be disqualified from that game.

5) Scoring:

- The contest will be based on a point system depending on the number of the wins.
- At the end of each game, each team will receive +2 point for each "assigned" colored box in their home locations and -1 points for each "other" colored boxes in their home locations. The team with the highest score will win the game and receive one win.
- **The ratio of assigned colored boxes versus other colored boxes is chosen so that if a team randomly collects all the boxes, it will receive a zero score ($+2*3 - 1*6 = 0$)!**
- The box must be completely inside of the home location area so to be counted.
- If boxes are contained in the body of a robot, then the most of the robot's body (>50) must be in the home location so as to be counted for the team.
- At the end of each game, **the winner robot** will also receive **3 game win points**.
- The boxes which are thrown outside of the play arena by a team will not be counted for/against any team.
- The **maximum** score can be received from each game is: $6 + 3 = 9$.
- The **minimum** score can be received from each game is: -6.
- A team may use different strategies such as collecting only the assigned colored boxes from the arena, stealing and moving out boxes from the opponent's home location, collecting other colored boxes and dumping them to the opponent's home location, carrying all assigned boxes in an internal storage, or delivering one box at a time, etc.
- **At the end of all games, the champion will receive extra 10 points for winning the tournament. The 2nd and 3rd place winners will receive 5 and 3 extra points, respectively.**
- Do not forget that your extra credit will depend on the total score that your team will receive from the tournament.

Final Report Guidelines:

Follow these guidelines closely to receive the full report-grade! The final project report will be important part of your grade since your grade will not be only based on the performance of your robot during the competition. The final project report must include these sections:

- **Title page (1%)**, including names of the team members, name of team's robot and date.
- **Objective Section (4%)**. This part explains the objective of the final contest in few sentences.
- **Robot Design (40%)**. This section must explain details of the design of team's robot, the reasons for the particular design selected, and pictures of the robot.
- **Robot Algorithm (45%)**. This section must explain the algorithm written in details, why the specific approach is chosen should be clearly stated. **Note that this section is not asking for the actual code.** It is asking the explanations of the code developed. The actual code should be emailed to the instructor.
- **Discussions and Conclusions (10%)**. This section should be written after the competition. What was the result of the competition? How well your algorithm and design of your robot performed? If it design poorly what are the reasons? How can they be improved? This section should also include the suggestions to improve the final competition.
- Note that even if your robot performs very poorly in the competition, you can still get a good report-grade by writing a nice report. The performance of your robot will not affect your report-grade.
- Final project report should not be less than 5 pages and more than 10 pages long. **Use 12-point Calibri font, single space.** Do **not** include your **code** to your report; instead email me your code.
- **A 10-page report full of pictures and code (again do **NOT** include your code) with minimal explanations, and discussions will receive a **very poor** report-grade.**

Grading

- The total grade of the robotic competition is 20% of your final course grade, i.e., 20 points, which will be distributed according to the following:
 - Final robot design 25%
 - Final robot algorithm 25%
 - Quality of the final project report 25%
 - Performance of your robot during the **mock** competition 25%
- **EXTRA CREDITS:**
 - It will be decided with the tournament number of games and/or elimination rules later.
 - Any negative extra point (!) will be rounded up to zero.

FINAL THOUGHTS:

- You are financially responsible for the robot kits that you have received.
- **Instructor has the right to change the rules, and/or grading policy for the competition as long as it is announced to the students in advance.**
- The TA spent a LOT of time this semester trying to organize the LEGO blocks and sensors. To respect the TA and next students, **you MUST return the equipment completely, in working condition, and as organized in the boxes as you received. Otherwise, you will receive NO credit from the course.**